TRANSPORTATION SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture with Mode Sub-Index

DBA-TRAN: Transportation Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

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For review by engineers, regulators, carriers, and policy analysts responsible for protecting transportation-sector critical infrastructure across the aviation, rail, maritime, and highway modes.

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Transportation Systems sector. It is the parent instrument for the DBA-TRAN series and governs the facility-class instruments of every type developed beneath it. It applies the DB → DBA → DBA-MA three-layer analytical progression: a baseline Design Basis establishing the postulated conditions the sector's facilities must withstand; a Design Basis Accident layer bounding the non-adversarial consequence envelope; and a Military-Action extension applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to the sector's governing postulated events.

1.2 What This Document Does Not Do

This document does not supersede or modify any modal safety or security regulation. It does not restate the governing definitions of the Bright Line or the Command Broker, inherited by reference from FD-BL. It does not re-scope infrastructure already governed by another series: fuel pipeline transport is governed by the Energy series (DBA-GAS-FC1 and DBA-OIL-FC4) and is cross-referenced here, not duplicated. It does not, at this version, develop the per-class analytical content, which the facility-class instruments named in the register carry.

1.3 Function-Primary Organizing Principle — and the Mode Transpose

The Transportation sector is conventionally organized by mode: aviation, rail, maritime, and highway, each with its own regulator, physics, and operating culture. That modal organization is descriptive and correct, but it is not the organizing axis a design basis uses. Design-basis placement follows the consequence chain a facility governs — the initiating events it must withstand and the mitigations available to answer them. Applying that principle, the sector decomposes by function into the nodes where movement aggregates, the ways along which it travels, the conveyances that carry it, and the control function that governs movement authority.

Mode is retained as the sub-index within each function class, in the same relationship that hazard class bears to the chemical function classes and fuel type bears to electric generation. This function-primary organization is the transpose of the conventional modal view: the modal taxonomy fixes the mode and enumerates its infrastructure, operations, and signaling; this framework fixes the function and enumerates its modes. Because the modes differ so profoundly — an air traffic control failure and a vessel traffic services failure are the same functional event in radically different physical settings — mode is a strong secondary axis, and several function classes are expected to resolve into mode-specific instruments early, just as electric generation resolved into fuel-specific facility classes.

1.4 Series Placement — DBA-TRAN as a Standalone Sector Series

DBA-TRAN is a standalone sector series, a peer of DBA-EN (Energy), DBA-W (Water), DBA-CHEM (Chemical), DBA-COMM (Communications), and the DBA-MA nuclear series. It is not subordinate to any other sector. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination.

1.5 Scope Boundaries and Cross-Series Interfaces

The Transportation sector touches more other sectors than any built so far, so its boundaries are stated explicitly. Pipeline transport of fuels is Energy, not Transportation, and is governed by DBA-GAS-FC1 and DBA-OIL-FC4. Pipelines that carry chemicals other than fuels, such as anhydrous ammonia, chlorine or ethylene, are governed by DBA-CHEM-TR-FC2 (Bulk Chemical Transport) on the author's decision of 5 October 2026, because their governing consequence is the chemical-release chain. Hazardous-materials rolling stock — the tank car and tank truck — is a boundary object shared with the Chemical sector: the chemical-release consequence chain is governed by DBA-CHEM and the transport-network consequence chain by DBA-TRAN, and both cite the shared chemical-release model. This closes the boundary rule registered as open in DBA-CHEM §3.3. Navigation locks and dams on the inland waterway system are a boundary object shared with the Dams sector; on the author's decision of 8 October 2026 the line runs by function, the lock chamber, its gates and valves and the machinery that operates them being DBA-TRAN-WY-FC2's and the dam, the impoundment and their release the Dams series' (Ways and Linehaul §1.2, §8.3). Postal and shipping logistics are noted as within the NIPP definition but not yet scoped.

Section 2 — The Transportation Sector and Its Regulatory Context

2.1 A Multi-Modal Sector Under the DOT Umbrella

The Transportation sector is governed by a family of modal administrations under the Department of Transportation, with security layered over them by the Transportation Security Administration and independent investigation by the National Transportation Safety Board. Aviation is governed by the Federal Aviation Administration; rail by the Federal Railroad Administration; highway and motor carrier by the Federal Highway Administration and the Federal Motor Carrier Safety Administration; maritime by the United States Coast Guard and the Maritime Administration, with the Army Corps of Engineers responsible for the inland waterway system and its locks. Hazardous-materials transport across all modes is governed by the Pipeline and Hazardous Materials Safety Administration. No unifying engineering design basis spans the modes.

2.2 Regulator-Per-Mode-and-Function Inheritance

Because both the mode and the function determine the applicable authority, the function-primary taxonomy with its mode sub-index is the natural structure for the regulatory terrain. The control

function inherits the modal traffic authority — the FAA for air traffic control, the FRA for positive train control, the USCG for vessel traffic services; nodes and ways inherit the modal infrastructure authority and, for security, the TSA; conveyances inherit the modal vehicle authority and, for hazardous cargo, the PHMSA hazmat regime that is shared with the Chemical sector.

2.3 The Regulatory Gap

Each mode carries a mature safety regime, but no instrument defines a cross-modal engineering design basis — the set of events transportation infrastructure must be demonstrated to withstand as a system, including the cross-modal and cross-sector cascades that no single modal administration owns. Security coverage is uneven: the TSA mandate is strong in aviation and comparatively light on the surface modes. The design basis constructed here is consequence-driven and cross-modal by design.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes and the Mode Sub-Index

The sector is decomposed into three function classes — nodes, ways, and conveyances — each realized as a physical facility class, and one coordination class, the traffic-management-and-control layer. Each is sub-indexed by mode. Each is placed by the consequence chain it governs and inherits its regulatory anchor from the intersection of its function and its mode.

3.2 Nodes / Terminals (ND)

The node function covers the fixed points where movement aggregates and transfers: airports, seaports and marine terminals, rail yards and intermodal facilities, transit stations, and major truck terminals. Its governing consequence is throughput loss at a concentration point, and its defining vulnerability is that a small number of high-throughput nodes — a major container port, a hub airport, a classification yard — carry a disproportionate share of national or regional flow, so their loss is a long-recovery, system-level event. On the author's decisions of 8 October 2026, recorded in DBA-TRAN-ND-FC1 (*Nodes and Terminals*): the node stops at its fence, its airspace, channels and approaches being the ways class's and its control facilities the coordination class's wherever they stand; it carries two consequence chains, throughput and occupancy, as each node's function requires, measured by stranded flow and clearance time; the loss of a platform that many nodes share is a design-basis event of its own; a node keeps its critical systems through the loss of any single supply point, and its power arrangements are made known to the utility that serves it; the release-and-network rule of Section 1.5 applies to petroleum, gas and chemical terminals and to hazardous cargo at the node; and where a transit station lies beneath a venue, the line with Commercial Facilities is drawn by who acts. The classes are written in the order the register numbers them, the coordination class last, on the author's decision of the same date, which settles the selection of the Phase-A instrument.

3.3 Ways / Linehaul (WY)

The way function covers the linear infrastructure along which movement travels: controlled airspace, track and roadbed, the inland waterway system and its navigation channels, highways and their major bridges and tunnels, and transit guideways. Its governing consequence is severance of a critical span. A single destroyed long-span bridge, a disabled lock on a commercially essential waterway, or a collapsed tunnel can sever a corridor with a recovery time measured in months or years — the transportation analog of the long-lead-component problem the energy series treats with the RT_HVLLC metric. On the author's decisions of 8 October 2026, recorded in DBA-TRAN-WY-FC2 (*Ways and Linehaul*): the line with the Dams series runs by function, as Section 1.5 states; airspace is a way in that class only where a physical hazard or a hostile threat denies it, and its routing, notices and separation are the coordination class's; a foreign maritime chokepoint is a condition the country imports and not a span the class designs; the governing measures are severance time, in its repair and replacement parts, and corridor stranded flow, replacement time being the class's counterpart of the long-lead recovery measure; and a span over navigable water carries both of the ways it severs, with the strike of a vessel a required event in its design basis.

3.4 Conveyances / Fleet (CV)

The conveyance function covers the mobile assets that carry passengers and cargo: aircraft, vessels, rolling stock, tank trucks, and transit vehicles. Two features distinguish it. First, the conveyance is the mode's point of highest kinetic energy and, for hazardous cargo, its point of highest release potential; a loaded toxic-inhalation-hazard tank car or tank truck is simultaneously a conveyance and a mobile chemical-release source. Second, that dual character is the formal boundary with the Chemical sector: the release consequence chain is governed by DBA-CHEM, the transport-network consequence chain by DBA-TRAN, and both cite the shared chemical-release model — the closure of the DBA-CHEM §3.3 boundary rule. On the author's decisions of 8 October 2026, recorded in DBA-TRAN-CV-FC3 (*Conveyances and Fleet*): at the tank car and the tank truck the line with DBA-CHEM runs between the container and the vehicle, the shell, its thermal protection, its relief device and the release being DBA-CHEM-TR-FC2's and the wheels, bearings, brakes and securement, and the car as a vehicle in its train, this class's; the wayside detectors that find a failing bearing, wheel or axle are this class's, which sets their spacing, thresholds and response; the governing measures are fleet unavailability, weighed by type concentration, and the control recovery margin, the time a conveyance takes to restore control against the time it has before contact; onboard autonomy is this class's and the authority to move the coordination class's; and the conveyance turned into a weapon is a design-basis event for the half the conveyance can answer.

3.5 CntlCo — Traffic Management, Signaling & Control (coordination class)

The control function is carried by a coordination class, spanning air traffic control, rail dispatch and positive train control, vessel traffic services, highway traffic-management centers, and transit

control. When a hazardous shipment is in trouble, the Command Broker role for decisions about its movement — routing, holding, speed, the release of a train — is held within this coordination class, and the shipping company owes its knowledge of the cargo to whoever decides about the cargo itself (DBA-CHEM-CorpCo, Chemical Enterprise Coordination and Control, on the author's decision of 5 October 2026). It is the movement-authority control plane — the layer that grants or withholds permission to move — and it is the transportation sector's closest analog to the electric sector's CntlCo. On the coordination spectrum the corpus has developed, transportation sits nearest electric: its control is a mandated, safety-critical, real-time authority, often federal (air traffic control most of all), in contrast to the communications sector's voluntary coordination commons (OpsCo) and the chemical sector's privatized coordination (CorpCo). It is also the locus at which the AI-Governance Design Basis is implemented: the Bright Line placement and Command-Broker obligation for automation in movement authority — autonomous separation, automated train control, autonomous vessels and vehicles — attach here. On the author's decisions of 8 October 2026, recorded in DBA-TRAN-CntlCo (*Traffic Management, Signaling and Control*): the governing measures are capacity restoration time and authority reach; the closure and reopening of a node or a way is split by legal authority, held in this class where the law gives a public authority the power and otherwise held by the owner on the basis this class supplies, a passenger node's closure staying with its operator; a wayside detector's alarm is enforced as a stop through train control, and only the dispatcher, on the crew's inspection, authorizes the train to move again; an autonomous conveyance receives authority to move only inside a pre-approved envelope matched to its certified operating conditions, and any grant above the line still requires a qualified human; and the governmental emergency acts on movement are recorded as acts outside the design basis.

† The DBA-TRAN-CntlCo designation deliberately reuses the CntlCo coordination-class name from the electric sector to record that transportation control is the same kind of mandated external real-time authority. The designation and the ND / WY / CV facility-class designators were ratified 2026-08-09.

3.6 Mode as the Sub-Index

Five modes sub-index the function classes: aviation; freight rail; passenger rail and transit; maritime and inland waterways; and highway and motor carrier. Pipeline, a sixth mode in the NIPP definition, is governed by the Energy series and carried here only by cross-reference. Postal and shipping is noted as within scope but not yet developed. As instruments mature, the highest-consequence mode-and-function intersections — the hub airport node, the ATC control class, the maritime chokepoint way, the hazmat conveyance — are expected to become the first mode-specific facility-class instruments.

Section 4 — The Transportation Sector Design Basis

The Transportation Sector Design Basis defines the envelope of events against which the sector's infrastructure must be shown to stand. The four class instruments develop it at class resolution, each for its own function, and this section states it once for the sector: the principles that govern

it, the principles the classes found and the framework adopts, the event categories across the four classes, the measures that size protection, and the critical functions the design basis requires. It cites the classes for what is particular to each and does not restate them.

4.1 Design Basis Principles

The criterion for inclusion is consequence severity — potential for loss of a movement capability on which public safety, national mobility, or another critical sector depends, at regional or national scale, or long-duration severance of a critical corridor — not event likelihood. Neither is attribution a criterion. An event is in the envelope if the path from its initiating condition to its consequence is reachable, whether the initiating condition is a failed bearing, a loose wire, a deleted file, an operator's error or an adversary's act.

Three further principles govern the envelope. First, the design basis is stated by function and differs by mode only where the mode changes what must be done; a requirement that names no mode applies to all five. Second, the sector's dependencies are part of the envelope: a node, a way, a fleet or a control facility may not claim to meet the design basis by relying on the continued availability of electric power, communications, or satellite positioning and timing whose loss is itself a design-basis condition (Section 6.2). Third, the people the sector holds and carries are within the envelope as well as its flow: the crowd in a terminal, the crew on a span, the passengers in a tunnel and aboard a conveyance are protected by requirements of the events that can harm them, stated in each class.

4.2 Sector Principles Adopted From the Classes

The four class instruments each found, from their own reference events, principles that hold for the whole sector. All four asked the framework to state the first of them, conditionality, for the sector, and the others recur across the classes. On the author's decisions of 8 October 2026 the framework adopts five. Each is stated here and applied by the classes; a class that applies one cites this section.

Conditionality. Any relaxation of a conservative class parameter must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. Each class lists the relaxations most likely in its domain (Nodes and Terminals §5.8; Ways and Linehaul §5.8; Conveyances and Fleet §5.8; Traffic Management, Signaling and Control §5.6).

Independence shown as built. Redundancy is credited by the independence of its parts as built, not by their number. A backup that shares a room, a duct, a fire zone, a route, a carrier, a database or an update channel with what it backs up is not independent of it. At Atlanta one fire took the airport's primary and backup supplies together (Nodes and Terminals §4.1); the backup of the notice system shared the deletion that took the primary (Traffic Management, Signaling and Control §4.3); and at Philadelphia the primary circuit and the backup fiber of the approach control facility were both lost on the same morning (Traffic Management, Signaling and Control

§4.2). Each class states the independence it requires — of supply points, of alternate routes, of the knowledge of occupancy from the logic that grants movement, of the paths that carry a facility's links — and an independence that is not shown is not credited.

The fallback is measured. Every fallback on which the design basis relies is a stated quantity, measured in exercises and in events, and not an aspiration: the node's manual fallback rate and hold time (Nodes and Terminals §3.4), the way's detour and the corridor stranded flow it leaves (Ways and Linehaul §3.3, §7.2), the conveyance's control recovery margin (Conveyances and Fleet §3.3), and the control plane's capacity restoration time (Traffic Management, Signaling and Control §3.1). A fallback that has never been measured is not known, and is not credited under the conditionality principle.

Common mode counted as one. What many units share is counted as one point of failure, and its loss as the loss of every unit that shares it: the software platform many nodes share (Nodes and Terminals §4.2), the type, the specification and the software build a fleet shares (Conveyances and Fleet §3.4, §4.2, §4.3), and the control facility, the circuit and the database on which a mode's movement authority depends (Traffic Management, Signaling and Control §2.1, §3.1, §4.2, §4.3). A change that reaches every unit at once — an update to a platform, a fleet or a control system — is a governed act (Section 5.4). Nagoya, the 737 MAX and the notice-system outage are the sector's references, with the A320 directive, withdrawn as a reference event, cited as an illustration (Conveyances and Fleet §2.3).

Act before it arrives. Where a hazard moves toward people or toward a structure, the time to warn, to clear and to restore control must each be shorter than the time the hazard takes to arrive, and each is stated and measured: the warning-to-clearance time of a span or tunnel (Ways and Linehaul §3.5), the control recovery margin of a conveyance in confined space (Conveyances and Fleet §3.3, §4.1), the clearance time of a node (Nodes and Terminals §3.3), and the time from a detector's alarm to the stop of the train (Conveyances and Fleet §4.4; Traffic Management, Signaling and Control §4.8). The Key Bridge is the sector's reference: the ship lost power too close to the bridge to restore it, and the warning stopped the traffic onto the span but did not reach the crew on it in time.

4.3 Event Categories

The envelope is organized by event category. Each category below names the postulated event at sector level, the consequence that places it in the envelope, the governing requirement, and the classes that develop it; the classes state the events, thresholds, required functions and durations in full at their Section 4.

Loss of supporting power. The loss of electric supply at a node, at the systems of a way — locks, movable bridges, tunnels, traction power — or at a control facility and its remote sites. Critical systems survive the loss of any single supply point, the switchover time is stated, and the power arrangements of a node are made known to the utility that serves it (Nodes and Terminals §4.1;

Ways and Linehaul §4.7; Traffic Management, Signaling and Control §4.1, §4.2, §8.2). Atlanta and Heathrow are the references.

Loss of shared software or shared information. The failure, compromise or corruption of a platform many nodes share, of the software a fleet runs, or of the information the control plane distributes. Every unit can work without it at a measured rate, sound software can be restored to every unit quickly, and a change that reaches every unit is a governed act (Nodes and Terminals §4.2; Conveyances and Fleet §4.3; Traffic Management, Signaling and Control §4.3). Nagoya, Maersk and the notice-system outage are the references; the Collins Aerospace attack and the A320 directive, withdrawn as reference events, are cited as illustrations (Nodes and Terminals §2.3; Conveyances and Fleet §2.3).

Natural-hazard exceedance and environmental denial. Hurricane, surge, flood, scour, earthquake and extreme heat beyond design, and the denial of a way without damage to it — low water, a canal without water for its locks, airspace closed by ash. Each structure states the hazards and return periods it is designed for, and environmental denial is counted as severance (Nodes and Terminals §4.3; Ways and Linehaul §4.4).

Severance of a critical span or passage. The loss of a long-span bridge, a lock, a tunnel or a channel, by vessel strike, structural failure, fire or obstruction, and the loss of a node’s access by the ways that feed it. A span over navigable water carries both of the ways it severs, the strike of a vessel is a required event in its design basis, and every way of consequence knows the replacement time of its critical spans before it needs it (Ways and Linehaul §4.1 to §4.6; Nodes and Terminals §4.5). The Key Bridge and Memphis are the references.

Loss of control of a conveyance. The loss of power, propulsion, steering or braking where the conveyance’s energy has little room, the failure of its running gear, and the failure of its securement when it is left unattended. Its control recovery margin is positive in each such place, its running gear is found failing before it fails, and its securement is counted without the brakes that can be lost (Conveyances and Fleet §4.1, §4.4, §4.5). The Dali, East Palestine and Lac-Mégantic are the references.

Fleet-wide defect. A defect in the design, manufacture or maintenance of a type, present in every unit, including a function that acts on a single indication it cannot know is false. A function that commands control acts on more than one independent indication, and the fleet unavailability of a grounding is known before it happens (Conveyances and Fleet §4.2). The 737 MAX is the reference.

Fire, explosion or release of hazardous cargo. A fire, explosion or release of hazardous cargo at a node, on a span, in a tunnel or aboard a conveyance. The release consequence is the series’ that holds the material, the transport-network consequence is this series’, and both use the shared release model (Section 1.5; Nodes and Terminals §4.4; Ways and Linehaul §4.3; Conveyances and Fleet §4.6). West, Graniteville and East Palestine are the references, conferred in the

Chemical series, West carrying the custody findings of the withdrawn Beirut instrument (DBA-CHEM-RE-5 F-11, F-12).

Crowd emergency and the weaponized conveyance. Fire, smoke, crush or the loss of systems in a full node, and an attack on the crowd, including by a conveyance driven into it. The node's occupancy can reach safety within its clearance time with its systems degraded, and a conveyance resists being taken and can be stopped when it is misused (Nodes and Terminals §4.7, §5.4; Conveyances and Fleet §5.4). Brussels, Nice, Berlin and New Orleans are the references.

Loss or corruption of the control plane. The loss of a control facility, its links, its shared information or its people, a wrong grant of movement by a person or by automation, and a command injected or spoofed. The control plane restores its capacity within its stated time, every proceed it grants is enforced by a layer whose basis is independent of the grant, and no channel delivers a proceed that has not been authenticated (Traffic Management, Signaling and Control §3.1, §4.1 to §4.7, §5.4). The Potomac, Philadelphia, Chatsworth, Fort Totten and the Polish radio-stop are the references.

Cyber attack. The compromise of the systems that run a node, operate a way, control a conveyance or grant movement, including through the suppliers and update channels that maintain them. The safety functions at every layer do not depend on the networked systems, the limits that protect equipment, structures and conveyances are enforced by the equipment itself, and the records and indications on which decisions rest are protected against alteration (Nodes and Terminals §4.8; Ways and Linehaul §4.8; Conveyances and Fleet §4.8; Traffic Management, Signaling and Control §4.6).

Loss or corruption of positioning, navigation and timing. Jamming or spoofing of satellite positioning and timing, a simultaneous multi-modal event (Section 6.2). Each element states what depends on it and what independent means remains, its timing receivers meet the design basis held in Positioning, Navigation and Timing §7, and a surveillance source that depends on satellite positioning is checked against one that does not before a grant of proceed rests on it alone (each class at §4.9). The interference in the Baltic region, DBA-COMM-RE-12, is the reference.

Flow beyond capacity and the imported chokepoint. Flow that exceeds what a node can clear, and the loss of a foreign chokepoint to armed conflict. Each node states its stranded flow, and each dependent corridor, node and sector states what the loss of a foreign passage costs it and what it holds to answer it, with a severance time set by the duration of the conflict (Nodes and Terminals §4.6; Ways and Linehaul §3.4). San Pedro Bay and the closure of the Red Sea and the Strait of Hormuz are the references.

4.4 The Sector's Measures

Each class sizes its protection by two measures, decided by the author on 8 October 2026: the node by stranded flow and clearance time (Nodes and Terminals §3.3), the way by severance time, in its repair and replacement parts, and corridor stranded flow (Ways and Linehaul §3.3), the conveyance by fleet unavailability, weighed by type concentration, and control recovery margin (Conveyances and Fleet §3.3), and the control class by capacity restoration time and authority reach (Traffic Management, Signaling and Control §3.1).

The measures compose. Each class's loss of function converts to flow stranded over a time: a node's flow that no other node can absorb while it recovers, a corridor's flow that no other way can carry while its span is rebuilt, the flow a grounded type would have carried, and the movement the control plane cannot grant until it is restored. Stranded flow over time is therefore the sector's common measure of its throughput harm, and it is the measure the forward-direction elevation of Section 6.1 uses: a transportation asset is the sole movement path for a dependent facility when the share of that facility's supply it would strand exceeds what the facility can hold in stock for the time it is stranded. The time-to-act measures — clearance time, warning-to-clearance time, control recovery margin and the detector-to-stop time — are the sector's measures of its harm to people, and they are held below the time the hazard takes to arrive (Section 4.2). Authority reach is the control class's measure of how much of both a single act can cause. No value is given here; each class states its values as open items at its Section 10.1, and the conditionality principle governs any value set below the class baseline.

4.5 Critical Function Summary

Across all event categories, five critical functions must be maintained or restored within the durations the classes specify.

The sector moves the flow on which other sectors and the nation's mobility depend, at no less than the stated fraction of its design flow, by the element itself or by an alternative planned with it, for the duration of the event.

The sector brings the people it holds and carries to safety: the occupants of a node within its clearance time, the people on a span or in a tunnel within its warning-to-clearance time, and the occupants of a conveyance by its means of escape.

Every conveyance keeps control of its energy, restoring its power, propulsion, steering or braking by its own means faster than the hazard arrives, and every unattended conveyance stays where it is left.

The control plane grants no proceed on a wrong or unauthenticated basis, and restores its capacity within its stated time.

The indications on which every decision rests — what a node holds, whether a span is sound, where a train is, what a ship is doing, what a sensor reads — are authentic, or are known to be suspect. Manual fallback, clearance and the authority to move are worthless against a falsified

picture, and the sector's records, indications and shared information are protected accordingly (Section 5.7).

Section 5 — Dual Design Basis: Bright Line and Command Broker

Transportation is entering the most consequential automation transition of any sector: autonomous aircraft separation, automated and autonomous train operation, autonomous vessels, and autonomous road vehicles all move machine systems into the movement-authority role, and machine-learning systems already plan yards, read inspection images, predict failures and propose flow. This section states the dual design basis once for the sector. The four classes name the actions their functions take and apply it (Nodes and Terminals §6; Ways and Linehaul §6; Conveyances and Fleet §6; Traffic Management, Signaling and Control §6).

5.1 The Two Design Bases

Every element of the sector carries two design bases at once. The physical design basis is the set of protective functions that keep its structures, equipment and conveyances within safe limits and its people safe, independently of the systems an attacker could reach, and each class defines its own: the power transfer, life-safety systems and equipment limits of the node; the interlocks, tunnel fire modes and closure gates of the way; the fail-safe brakes, power restoration, emergency braking and envelope protections of the conveyance; and the vital logic and train-control enforcement of the control plane. Those functions are to be locally sovereign as FD-LS (Local Sovereignty) defines the property, and each class says which of its functions must be (Nodes and Terminals §5.6; Ways and Linehaul §5.6; Conveyances and Fleet §5.6; Traffic Management, Signaling and Control §5.4). The AI-governance design basis is governed by the Foundational Definitions and the Command Broker family and is named and referenced here.

Governing instruments. This framework does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (The Bright Line), which governs. The Broker role that discharges that obligation is defined in FD-BR (The Broker). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (Layer Decomposition and Determinism). Two determinations bind under FD-BL: FD-BL-D1 (Mode of Discharge) and FD-BL-D2 (Indication Integrity). FD-EV (The Envelope) reserves the term "envelope" for the pre-authorized envelope within which advance discharge is valid. The cross-sector instrument for the Command Broker architecture is CB-Framework_Cross_Sector (Command Broker Framework). The Foundational Definitions are cited without a version token; FD-REG (Governing Stack Register) is authoritative for their version state, and CB-REG (Command Broker Family Register) for the Command Broker family's.

5.2 Placement by Action, in Four Places

Placement is on the consequence of the individual action, assessed per action (The Bright Line §4.2, §4.3); this framework applies that definition and does not vary it. Determinism decides scope under FD-LD §4.3; it does not decide placement. A component that is deterministic within a stated range of inputs and formally tested as installed is outside the Command Broker obligation for the actions it takes while its inputs stay in that range, and its actions are not thereby placed on either side of the Bright Line (The Bright Line §4.5). The range must include the inputs the component can meet in service: a function that can receive a false indication it cannot detect, as the 737 MAX's flight-control function did, or that can lose detection of what it protects, as Fort Totten's train control did, has not been shown to meet the test for those inputs.

The actions to which the architecture applies arise in four places, one for each class, and each class names them. At the node they are stowage and yard plans that place hazardous cargo, the moves of automated cranes and vehicles where people may be, crowd actions while the node is full, and the decision to close or reopen a passenger node (Nodes and Terminals §6.3). On the way they are the classification of an inspection finding on a member without an alternative load path, the movement of a lock gate or a movable span, the setting of a tunnel's ventilation in fire, and the decision to close or reopen a critical span or channel (Ways and Linehaul §6.3). Aboard the conveyance they are the driving actions of an automated vehicle, the return to service of a unit whose failure could kill, the update of the software that commands control across a fleet, and the decision to ground a fleet or keep it in service (Conveyances and Fleet §6.3). And in the control plane they are the grants of movement authority (Traffic Management, Signaling and Control §6.4). For screening at a node, a second Line governs the liberty consequence of the same action (Section 5.8).

5.3 The Proceed Is the Placed Act

On the author's decision of 8 October 2026, the sector states one placement principle for movement authority. An automated grant of movement — a clearance, a route, a block, a speed command, a direction — is above the line where a wrong grant could bring a movement into conflict with another, or with something on the way, at or above the severity threshold, and it requires the architecturally enforced authorization of a qualified Command Broker, given in the moment or in advance under Section 5.4. A stop is the fail-safe direction, and a command that can only slow or stop a movement is placed on its own consequence, which is the flow it strands and the conflicts a sudden stop can cause. The principle names which actions of a control system carry the collision consequence; placement remains per action. Its references are Fort Totten, where automatic train control granted speed to a train with another stopped ahead, and the Polish radio-stop, where an injected command could only stop (Traffic Management, Signaling and Control §2.3, §4.5, §4.6, §6.4).

5.4 Advance Discharge in the Sector

Most of the sector's automated actions are repeated continuously within limits set in advance — a crane's moves, a lock's routine operation, an automated vehicle's ordinary driving, automatic

train control's speed commands, the sequencing of arrivals — and the obligation for them may be discharged in advance under Mode of Discharge, within a pre-approved envelope (Pre-Approved Action Envelope §4) whose limits the responsible owner sets and whose gatekeeper is verified. Three conditions apply across the sector. The gatekeeper is verified, and for movement authority, on the author's decision of 8 October 2026, advance discharge is valid only inside an envelope whose gatekeeper knows the occupancy of the space independently of the logic that grants movement into it (Traffic Management, Signaling and Control §6.4). A change to the limits of an envelope, including by a change to software or data, is a change to the envelope and is excluded from advance discharge (Mode of Discharge §4.3). And an action that reaches every unit of a platform, a fleet or a control system at once is one action, whose consequence is the sum of its consequence on every unit it reaches; it is a governed act, authorized by a qualified person, and no automated pipeline sends it alone (Conveyances and Fleet §6.3).

5.5 Who Holds the Command Broker Role

There is one Command Broker role (The Broker §4.2), held by named, qualified people for different actions, and each control facility records its holders in an authorization map (Traffic Management, Signaling and Control §3.3, §6.2). The sector places the role as follows. For movement through controlled space it is held by the controlling authority's qualified people — the controllers, dispatchers and watchstanders — and the operator of a conveyance keeps its authority over its own conveyance, including its emergency authority, without holding the authority to grant itself movement into controlled space (Traffic Management, Signaling and Control §6.3). For closures it is split by legal authority: where the law gives a public authority the power to close a node or a way, the control class holds it; otherwise the owner of the node or way decides on the basis the control class supplies, and the closure of a passenger node stays with its operator (Traffic Management, Signaling and Control §6.3; Nodes and Terminals §6.3; Ways and Linehaul §6.3). When a hazardous shipment is in trouble, the role for decisions about its movement is held in the control class, and the shipper owes its knowledge of the cargo to whoever decides about the cargo itself (Section 3.5). For the grounding of a fleet the role is held by the operator and the regulator, each under its own authority (Conveyances and Fleet §6.3). The authorities of the states of design, registry and operation each hold the decision to ground a type or return it to service, and each records the basis on which it acts under Indication Integrity; where one manufacturer's assessment is every authority's basis, that assessment is the basis that carries Indication Integrity, and the order of the state of design is the act the others adopt, each on its own terms (Boeing 737 MAX F-6, which carries the finding of the withdrawn Airbus A320 directive instrument). Where a holder's interest in throughput runs against refusal, the holder must be able to refuse, or to ask for a lower rate, without answering to the person who gains, and the request is recorded and answered (Traffic Management, Signaling and Control §6.7).

5.6 Autonomous Conveyances

On the author's decisions of 8 October 2026, the system that drives a vehicle, navigates a vessel or flies an aircraft in place of a person is designed in the conveyance class, and the authority to move it is granted by the control class (Conveyances and Fleet §4.7; Traffic Management, Signaling and Control §6.5). An autonomous conveyance receives authority to move only inside a pre-approved envelope matched to the operating conditions for which its own regulator certifies it, and any grant of authority above the line outside that envelope still requires a qualified human holder. A machine may receive authority; no machine grants above-line authority alone. An autonomous conveyance that loses its link to the person who supervises it takes no above-line action it could not have taken with the link, and stops safely instead (Conveyances and Fleet §6.3; Military Action Communications Design Basis, CDB-6). The highway mode has no function that grants movement authority, and the state permits under which automated vehicles operate are outside this design basis, an open item for the author (Traffic Management, Signaling and Control §10.6).

5.7 What a Supplier Declares, and the Integrity of the Basis

Whether the authorization obligation reaches an action depends on how the system is built. Where a person reads a system's output and then decides and acts, the action is the person's, and the output carries the obligation of Indication Integrity rather than the authorization obligation (Mode of Discharge §7). Where the output itself makes the change — a crane that moves when it is planned, a gate that closes when a model says so, a vehicle that moves when its planner says so, a grant issued by a system — the authorization obligation reaches it (The Bright Line §4.3; Mode of Discharge §7). Every supplier to the sector declares, for each output of its product, which of the two it is, and the vendor's description does not govern; the declaration and the architecture do. For a conveyance or control system with no person in a position to read the output, every action is of the second kind.

The basis on which an authorization rests must satisfy Indication Integrity. In this sector the basis is very often a single indication carried far from its source: a container's location in a terminal operating system, a manifest, an inspection reading, a sensor on a flight-control computer, a vessel's broadcast position, a track circuit's report that a block is clear. Each class requires that the indications its decisions rest on be protected against alteration (Nodes and Terminals §4.8, §5.5; Ways and Linehaul §4.8; Traffic Management, Signaling and Control §4.3, §4.6). A function that commands a conveyance's control acts on more than one independent indication and stops acting when they disagree (Conveyances and Fleet §4.2), and a surveillance source that depends on satellite positioning is checked against one that does not before a grant of proceed rests on it alone (Traffic Management, Signaling and Control §4.9).

5.8 Two Lines, and the Acts Outside the Design Basis

Some actions at a node carry a liberty consequence as well as a security consequence. Passenger screening is the case: Nodes and Terminals §6.4 names FD-BL for its security consequence and

FD-SBL (The Surveillance Bright Line) for its liberty consequence, as FD-SBL requires of a document that sits in both corpora, and an action above both Lines takes both obligations.

The governmental emergency acts on movement — the emergency security control of air traffic, a national ground stop, a Captain of the Port's emergency order — are recorded as governmental acts outside this design basis, as the communications series records the President's emergency powers over communications (Traffic Management, Signaling and Control §6.6; Operations and Coordination §6.4). They do not replace the controlling authority's holder, who still authorizes the execution of each act in the space it controls on a basis whose integrity Indication Integrity governs, and the design basis requires that each control facility be able to receive such an order over a channel it can trust, know its authenticity before it acts on it, and carry it out (Traffic Management, Signaling and Control §6.6).

5.9 Continuity

Consistent with CDB-6 of the Military Action Communications Design Basis, the loss or degradation of an element's communications does not suspend the Command Broker obligation. CDB-6 states the obligation for actions recommended by generative AI; this framework applies it to every component within the scope of Layer Decomposition and Determinism §4.3, which is the scope the Foundational Definitions set. In every class the control system must be architecturally incapable of executing an above-line action originating from a component within that scope without Command Broker authorization, and that holds whether the component sits at a node, on a way, aboard a conveyance or at a control facility.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Outward View — Transportation as the Supply-Chain Enabler

Transportation is the sector through which every other sector's physical goods move. Its outward consequence is therefore a supply-chain consequence: severance of a transportation node or way curtails the delivery of fuel, chemicals, food, and components to the sectors that depend on them, and the forward-direction elevation obligation applies where a transportation asset is the sole movement path for a Tier-2 or Tier-3 facility in another series — the rail line that is the sole coal or chemical supply route, the port that is the sole fuel-import terminal. Transportation also carries the military-mobility consequence: ports of embarkation, strategic sealift, and the rail and highway networks that move force are national-security assets whose loss is a strategic, not merely economic, event.

6.2 Inward View — Energy, Communications, and PNT

Transportation's reverse dependencies run to three sectors. It depends on Energy for motive fuel (jet fuel, diesel, marine fuel) and for the electric power that runs rail electrification, transit, signaling, and air traffic control. It depends on Communications for the control-plane links of

Section 3.5 — air-ground communications, dispatch, and the data paths of positive train control and vessel traffic services. And it depends acutely on position, navigation, and timing: aviation, maritime, and rail navigation all rest on the GNSS timing that the DBA-COMM framework identifies as its most consequential common-cause failure. A PNT denial is a simultaneous multi-modal transportation event, and this dependency must be stated as a design basis condition.

Regional interference with satellite navigation is such a condition, and it is already standing. Interference across the Baltic region has continued from 2023, degrading satellite navigation for aircraft and ships across the region and the time of spoofed aircraft; Tartu's airport suspended flights from 29 April to 31 May 2024, after two flights turned back, until a ground-based approach aid came into use, the dependence of its approaches on satellite navigation being Finnair's account (Baltic Navigation Interference (DBA-COMM-RE-12) F-7, F-9; Positioning, Navigation and Timing §2.3), Finnair noting the airport's instrument landing system for one direction. The series states its consequence for aviation and maritime navigation: a node whose usable approaches depend on satellite navigation, a way whose passage does, and a conveyance that navigates by it each state the stranded flow with satellite navigation denied for as long as the interference can be sustained, which in the Baltic has been years, and the alternative means that restore the approach or the passage. The timing function of every receiver is held at Positioning, Navigation and Timing §7, whose §7.3 draws, for the author's ruling, the line between it and the navigation function aboard aircraft and ships. The classes conform at their next revisions.

What a node or a way requires of the electric supply that serves it, including a supply another party runs inside its fence, is stated in its own design basis and reaches the utility as quality of service, through an agreement or tariff that names the event the supply must survive. A statistical service level does not discharge it, because how often and how long customers lose supply is a measure of likelihood, and the inclusion criterion is consequence (Nodes and Terminals §4.1, §8.3; Ways and Linehaul §8.3). The water series' elevation rule, which raises the acceptance criteria of a supplying facility for dependents such as dialysis, fire flow and cooling water, is not applied here, because a node or a way can contract for its supply and those dependents generally cannot (Nodes and Terminals §8.3). On the author's decision of 10 October 2026 the question first put to the electric series under Project 9 OI-27 is withdrawn, and who bears the cost of that service is Project 9 OI-17.

6.3 Boundary Objects

Three boundary objects are governed jointly with other series and are named here so the interface is unambiguous: hazardous-materials rolling stock and tank trucks with the Chemical sector (Section 3.4); navigation locks and dams with the Dams sector; and pipeline transport with the Energy series. In each case the transport-network consequence is DBA-TRAN's and the other consequence chain is the partner series', with a shared model cited by both. For navigation locks the line runs by function (Ways and Linehaul §8.3). For hazardous-materials rolling stock the line runs between the container and the vehicle (Conveyances and Fleet §8.3).

Section 7 — The Military-Action Extension

Transportation is a primary military objective in its own right: it is how force projects and how a society sustains itself under stress. This section states the military-action extension once for the sector. The four classes apply it to their functions at their Section 5, and each develops a governing scenario and an adequacy condition of its own (Nodes and Terminals §5; Ways and Linehaul §5; Conveyances and Fleet §5; Traffic Management, Signaling and Control §5).

7.1 MA Extension Applicability Determination

Application of the extension is a mandatory determination for every facility, way, fleet and control facility of consequence, made under the four-filter applicability gate of DBA-MA-AF (MA Applicability Framework) §2 and §3, in which any one filter satisfied brings the element into scope. For a conveyance the determination is made at the level of the fleet as well as the unit (Conveyances and Fleet §5.1). Strategic value in this sector is driven by throughput concentration, sole-path status, the replacement time of what would be lost, and the element's role in moving the nation's forces.

On the author's decisions of 8 October 2026, the determination is presumed positive for five groups. The first is the designated strategic seaports and the nodes whose flow has no absorbing alternative (Nodes and Terminals §5.1). The second is the highways of the Strategic Highway Network with their connectors to installations and ports, the rail lines of the Strategic Rail Corridor Network, the channels and approaches of the strategic seaports, and the ways whose flow has no absorbing alternative (Ways and Linehaul §5.1). The third is the foreign maritime chokepoints, as conditions the country imports (Ways and Linehaul §3.4, §5.1). The fourth is the mobilizable fleet: the aircraft committed to the Civil Reserve Air Fleet and the ships of the Ready Reserve Force and of the Maritime Security and Tanker Security Programs (Conveyances and Fleet §5.1). The fifth is the key control facilities: the Air Traffic Control System Command Center, the air route traffic control centers and the major approach control facilities, the dispatch centers and train control back offices of the Class I and major passenger railroads, and the vessel traffic services at the strategic seaports (Traffic Management, Signaling and Control §5.1). Every other element runs the ordinary determination. A presumption can be rebutted only on a recorded basis, under the conditionality principle (Section 4.2). Military installations, military aircraft and vessels, and military air traffic control are outside the sector; a civil element designated for military use is inside it and carries the presumption.

7.2 Why the Non-Adversarial Design Basis Is Insufficient

Each bounding event of Section 4 shares a characteristic that distinguishes it from military action: it is not chosen. A wire works loose, a bearing overheats, a contractor deletes the wrong files, a drought lowers a lake. The sector's own history shows what changes when the event is chosen. The Houthi campaign has held the Red Sea impaired for almost three years, and the closure of the Strait of Hormuz in 2026 had nothing to do with the strait's condition; the Kerch

Strait bridge was brought down in part in 2022 and again in 2023, and restored each time by a state that treated it as strategic, and by Ukraine's account was struck a third time in 2025; and the Polish radio-stop showed that a control channel can be reached with equipment anyone can buy. The extension builds on the non-adversarial design basis and does not replace it: each bounding event has a military-action analog, deliberately caused, precisely aimed, timed to the moment it matters most, and executed to defeat the recovery on which the non-adversarial design basis relies.

7.3 The Four Characteristics Applied to the Sector

Intentionality. The adversary chooses the element whose loss strands the most flow, holds the most people, severs the corridor with the longest replacement time, grounds the largest fleet, or stops the most movement at once, and times the act to a mobilization, a crisis or the hour of greatest exposure.

Persistence. The adversary sits inside the platforms, fleets and control systems that many elements share, and inside the suppliers and update channels that maintain them, long enough to learn them and to reach every unit at once; it returns to a span as it is repaired; and it holds a passage closed for as long as the threat to ships can be sustained.

Precision. The adversary takes the one supply point, platform, span, sensor, circuit or channel on which an element depends, and leaves the rest alone, because the rest does not matter.

Denial. The adversary takes the element and its alternative together — the node and the port that would take its flow, the bridge and the detour, the fleet and its reserve, the control facility and its alternate — and corrupts the indications on which recovery depends, so that the flow has nowhere to go and what remains cannot be trusted.

The validation standard for the sector is the state actor who knows that transportation is where a nation's ability to move its forces and sustain its people can be stopped with the fewest acts.

7.4 The Governing Campaign

The four classes' governing scenarios are one campaign seen from four places. Forces deploying to a theater move from their installations by the designated highways and rail lines to the strategic seaports and airfields, and leave by sealift and airlift through channels and airspace under the control plane's authority. A campaign aimed at that deployment denies the strategic seaport by cyber attack on its terminal systems, by cutting its power and by closing or threatening its channel (Nodes and Terminals §5.3); severs the deployment corridor at its critical span and its nearest alternative together (Ways and Linehaul §5.3); strikes the mobilizable fleet at activation, through its maintenance and update channels, its positioning and its old machinery (Conveyances and Fleet §5.3); and denies the control plane by taking a control facility and its alternate and injecting false commands into the channels that remain (Traffic Management, Signaling and Control §5.3).

The sector's adequacy test is the conjunction of the four classes' adequacy conditions: the Sustained Node Condition, the Sustained Corridor Condition, the Sustained Fleet Condition and the Sustained Control Condition (Nodes and Terminals §5.6; Ways and Linehaul §5.6; Conveyances and Fleet §5.6; Traffic Management, Signaling and Control §5.4). For the design duration of the campaign, the military-mobility flow moves at a stated rate through the port or its planned alternative, along the corridor or its planned alternative, in the share of the mobilizable fleet ready within its window, under the grants of the control plane or its alternate, with the elements an adversary would choose lost in each. The analysis is made for the campaign as a whole and not class by class, because a plan in which each class names the others' alternatives can count one alternative several times; under the conditionality principle an alternative is counted once, against the same campaign.

7.5 The Imported Condition

Two conditions the country cannot design reach every corridor that depends on them. The first is the foreign maritime chokepoint — the Suez Canal and the Bab el-Mandeb, the Strait of Hormuz, the Panama Canal, the Strait of Malacca — whose loss to armed conflict is a condition the country imports (Ways and Linehaul §3.4, §5.4). The second is foreign airspace closed by conflict, as Russian airspace has been closed to most European and North American carriers since 2022, and as the airspace of several Middle Eastern states was closed from 28 February 2026, the European Union Aviation Safety Agency's conflict-zone information bulletin CZIB-2026-03 running to 8 July 2026 and its CZIB-2026-07, for the Persian Gulf and the Gulf of Oman, valid to 16 November 2026. Such a closure lengthens routes and reroutes the airlift that would cross it. For both, the design basis requires each dependent corridor, node, fleet and sector to state the stranded flow if the passage were closed for the length of a conflict rather than the length of a repair, the diversion it would take with the added time and the ships or aircraft it would need, and the stocks it holds against the longer voyage, with the military-mobility flow stated separately. An enclosed sea closed by conflict also holds what is inside it: about 20,000 seafarers and 850 to 1,000 ships were held in or off the Gulf, five of them United States-flag, on the counts Conveyances and Fleet §3.3 attributes (Red Sea and Hormuz Closure (DBA-TRAN-RE-10) F-5, referred to the framework by Project 11). Each dependent fleet therefore also states the people and conveyances it would have inside such a sea and the means of bringing them out.

7.6 The Design Duration Is a Campaign

On the author's decision of 8 October 2026, the framework adopts for the sector the principle the Emergency Services series adopted from two of its classes (Emergency Services Sector Framework §7): under military-action conditions the design duration is a sustained campaign as well as an event. Capacity is counted with the replacement of what the campaign consumes — the crews and controllers who would work by hand or fall ill, the spares for locks and cranes, the members for a damaged span, the circuits for a control facility, the ships and aircraft lost or damaged — and not only with the stock on hand when the campaign begins. A change to a

system in use, including a change made to restore it, is a governed act exercised before it is relied on.

The number of days is not set here. The four classes each left it to be set with the energy series' assumptions about restoration after a deliberate attack on the grid and the communications series' design outage, so that the series assume the same event (Nodes and Terminals §10.2; Ways and Linehaul §10.2; Conveyances and Fleet §10.2; Traffic Management, Signaling and Control §10.2). That is a decision across series, and it is registered as an open item for the author (Section 9).

7.7 The Declared Residuals

No design basis makes the sector proof against a state that sets out to stop it, and the classes declare what remains, in the output form of DBA-MA-DiD (Defense in Depth) §3.5. Taken together, the sector's residuals are of four kinds. The first is flow stranded while an element works below its design rate — the node by hand, the corridor by detour, the fleet short of units, the control plane from its alternate — bounded by the measured fallback and by the stocks the dependent sectors hold. The second is the loss of an element and its alternative together, which no measure inside either restores: for the node and the way the answer lies with the national response, and for the control plane the stranded movement until one is restored is accepted by the responsible authority. The third is what cannot be replaced quickly whatever is prepared: a fallen long-span bridge, an old and small sealift fleet. The fourth is what the country does not control at all: a foreign chokepoint, foreign airspace, and the carriers whose networks carry the control plane's links. Where a residual is accepted rather than answered elsewhere, each class records the decision to accept it as the responsible authority's, in the element's plan (Nodes and Terminals §5.7; Ways and Linehaul §5.7; Conveyances and Fleet §5.7; Traffic Management, Signaling and Control §5.5).

7.8 Re-Determination and Re-Validation

A determination that defers the extension is not permanent. It is revisited on a cycle the element's regulatory regime sets, and at once on any of the following: the mobilization of forces whose deployment the element would carry; the escalation of a conflict that could involve the country or the passages its trade and its sealift depend on; a credible threat by a state or state-sponsored actor against transportation infrastructure; or a confirmed attack on an element of the same class in a comparable setting, as the attacks on shipping in the Red Sea, on the Kerch Strait bridge and on the Polish railway's radio were. A deferral that was valid before such an indicator is not valid after it until the determination is made again.

For an element whose determination is positive, the criteria of its design basis are re-validated against observed adversary practice at an interval the element's plan states, until the classes set one, and a material change in that practice — a new means of reaching a control channel, a new kind of attack on ships, a new use of a conveyance as a weapon — triggers re-validation, as the

Emergency Services series requires for its own criteria (Emergency Services Sector Framework §7).

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-TRAN series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework. Each instrument adds what is specific to its function and mode — its bounding events, consequence chains, design parameters, and recovery architecture.

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-TRAN series. All four classes — DBA-TRAN-ND-FC1 (Nodes and Terminals), DBA-TRAN-WY-FC2 (Ways and Linehaul), DBA-TRAN-CV-FC3 (Conveyances and Fleet) and DBA-TRAN-CntlCo (Traffic Management, Signaling and Control) — are at working-draft status, all issued in October 2026; pipeline is shown as governed elsewhere and postal/shipping as not yet scoped. Designators were ratified 2026-08-09, and each physical class is sub-indexed by mode.

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory anchor
DBA-TRAN-ND-FC1	Nodes / terminals — airports, seaports & marine terminals, rail yards & intermodal, transit stations, truck terminals	concentration & throughput chokepoints; mode-sub-indexed	Working draft	FAA; USCG / MARAD; FRA; FTA; TSA (security)
DBA-TRAN-WY-FC2	Ways / linehaul — airspace, track & roadbed, inland waterways & channels, highways & major bridges/tunnels, transit guideways	severance of a critical span (bridge, lock, chokepoint)	Working draft	FAA; FRA; FHWA; USCG; USACE (waterways / locks)
DBA-TRAN-CV-FC3	Conveyances / fleet — aircraft, vessels, rolling stock (incl. TIH tank cars), tank trucks, transit vehicles	weaponization / hazmat release; boundary with DBA-CHEM	Working draft	FAA; USCG; FRA; FMCSA; PHMSA (hazmat)
DBA-TRAN-CntlCo†	Traffic management, signaling & control (coordination class) — ATC, rail dispatch & PTC, vessel traffic services, highway TMCs, transit control	the movement-authority control plane	Working draft	FAA (ATC); FRA (PTC); USCG (VTS); state DOT
(cross-ref)	Pipeline transport — governed by the Energy series (DBA-GAS-FC1, DBA-OIL-FC4); not re-scoped here	—	Governed elsewhere	PHMSA; TSA (pipeline security)
(noted)	Postal & shipping / parcel logistics	—	Not yet scoped	USPS; PHMSA (hazmat parcels)

Designators were ratified 2026-08-09 at v0.2; the designators of all four classes now name issued instruments. Facility-class numbers are sequential across the series; CntlCo is a coordination class and carries no FC number, mirroring DBA-ES-CntlCo. Each physical class (ND, WY, CV) is sub-indexed by mode (aviation, freight rail, passenger rail & transit, maritime, highway).

Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-TRAN sector framework. It establishes the standalone-series placement, the function-primary decomposition (nodes, ways, conveyances, and the CntlCo control class) with mode as the sub-index, the transpose relationship to the conventional modal view, the multi-modal regulatory terrain, and the cross-sector elevation structure — closing the DBA-CHEM §3.3 hazmat boundary and stating the Dams and Energy interfaces. The design-basis envelope (Section 4), the dual design basis (Section 5) and the military-action extension (Section 7) are developed at version 0.10 from the four classes, on the author's decisions of 8 October 2026. Open items at this version: the shared chemical-release model is now held, provisionally, at DBA-CHEM-PR-FC1 (Chemical Production and Synthesis) Section 4 and is cited from DBA-CHEM-TR-FC2 (Bulk Chemical Transport), which adopts this framework's boundary rule; the division of Command Broker authority between carrier and shipper is confirmed on the author's decision of 5 October 2026 and stated at Section 3.5 (OI-19 closed); and the Phase-A selection is settled: on the author's decision of 8 October 2026 the classes are written in register order, and all four classes are issued, developing class-level content ahead of Sections 4, 5 and 7: DBA-TRAN-ND-FC1 (Nodes and Terminals), DBA-TRAN-WY-FC2 (Ways and Linehaul), DBA-TRAN-CV-FC3 (Conveyances and Fleet) and DBA-TRAN-CntlCo (Traffic Management, Signaling and Control). Section 4.2 adopts for the sector the consequence-basis conditionality principle all four classes state (Nodes and Terminals §5.8, §10.1; Ways and Linehaul §5.8, §10.1; Conveyances and Fleet §5.8, §10.1; Traffic Management, Signaling and Control §5.6, §10.1), with four further principles. Whether the release of a fuel carried by rail or road is governed by DBA-CHEM-TR-FC2 or by the Energy series, which Section 1.5 does not say, is open for the author (Conveyances and Fleet §10.4). Open at this version: the number of days of the sector's military-action design duration, which Section 7.6 leaves to a decision across series with the energy and communications series. The six node instruments and the three ways instruments return elements to this framework, recorded by the classes (Nodes and Terminals §10.5; Ways and Linehaul §10.7), among them Key Bridge Collapse R-1 for Section 4.3, Key Bridge Collapse R-9, Hernando de Soto Bridge Fracture R-2 and Red Sea and Hormuz Closure R-6 for Section 4.2, and Red Sea and Hormuz Closure R-7 for Section 4.5; they are taken in at a conformance pass of this framework. On the author's decision of 10 October 2026 the material the reference events offered for use cases is folded into the existing use cases at their next revisions, and no new transportation use case is opened.

9.1 Conferred Reference Events

Under the author's decision of 6 October 2026 that every reference event a class proposes is conferred, this framework confers a reference-event designator on each of the events its classes propose. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event on the model of the water series' DBA-W-RE-1 (Minnesota Water Campaign) and DBA-W-RE-2 (Oldsmar Water Treatment). The designators are numbered in the order the proposing classes are

issued and, within each class, in the order the class lists its events. The number records conferral, not issue. Each designator is sector-primary and single: an event whose consequence reached other sectors is conferred in the sector that proposed it, and its reach is recorded in the register's impacted-sector column. Under the determination of 9 October 2026 (Reference Events Move to Their Own Project), reference-event instruments are held by Project 11, DBA Reference Events, and this framework keeps the conferral; an instrument not yet moved stays in Project 9's folders until it moves. On the author's rulings of 9 and 10 October 2026 two conferrals are withdrawn and one recast, and their designators retired and not reused.

From Nodes and Terminals, seven events are conferred: DBA-TRAN-RE-1, the NotPetya attack on A.P. Moller-Maersk's container terminals of June 2017; DBA-TRAN-RE-2, the ransomware attack on the Port of Nagoya's terminal system of July 2023; DBA-TRAN-RE-3, the power outage at Hartsfield-Jackson Atlanta International Airport of 17 December 2017; DBA-TRAN-RE-4, the closure of Heathrow Airport after the North Hyde substation fire of March 2025; DBA-TRAN-RE-5, the bombings at Brussels Airport and Maalbeek metro station of 22 March 2016; DBA-TRAN-RE-6, the congestion of the San Pedro Bay ports from 2020 to 2022; and DBA-TRAN-RE-7, the ransomware attack on the Collins Aerospace check-in and boarding system of September 2025. All seven instruments were written. DBA-TRAN-RE-7 is withdrawn on the author's triage ruling of 9 October 2026 and its designator retired, nothing being carried, because its shared-platform findings are held by DBA-TRAN-RE-1 and DBA-TRAN-RE-2; Nodes and Terminals cites the event as an illustration. The six live instruments are held by Project 11, and Nodes and Terminals took in what they returned at its v0.2. The class cited the explosion in the Port of Beirut, DBA-CHEM-RE-6, which is withdrawn; it now cites DBA-CHEM-RE-5 (The West Fertilizer Explosion), which carries Beirut's findings.

From Ways and Linehaul, four events are conferred, on the author's choice of 8 October 2026: DBA-TRAN-RE-8, the collapse of the Francis Scott Key Bridge at Baltimore of 26 March 2024; DBA-TRAN-RE-9, the fracture found in the Hernando de Soto Bridge at Memphis of May 2021; DBA-TRAN-RE-10, the closure of the Red Sea and the Strait of Hormuz to much of the world's shipping from November 2023; and DBA-TRAN-RE-11, the attacks on the Kerch Strait bridge of October 2022 and July 2023. The third joins two passages in one designator because the class reads them as one condition; on the author's ruling of 10 October 2026 they keep one designator, the coupling between them being the design-relevant fact, and the Red Sea half does not meet the state-actor validation standard in its own right, the event meeting it through the Hormuz half. All four instruments were written. DBA-TRAN-RE-11 is recast on the author's rulings of 9 and 10 October 2026: it is not a reference event, carries no tier, and is an illustration within Ways and Linehaul §5.5, its restoration record read as evidence of replacement time; its F-2 and F-4 are carried by DBA-TRAN-RE-8 F-9, and its designator is retired. The question it carried is answered generally by the author's ruling of 10 October 2026 that the military-action threshold is not met without denial. The three live instruments are held by Project 11, and Ways and Linehaul took in what they returned at its v0.2.

From Conveyances and Fleet, three events are conferred, on the author's choice of 8 October 2026: DBA-TRAN-RE-12, the crashes of Lion Air Flight 610 and Ethiopian Airlines Flight 302 of October 2018 and March 2019 and the grounding of the Boeing 737 MAX that followed; DBA-TRAN-RE-13, the runaway and derailment of a crude-oil train at Lac-Mégantic, Quebec, of 6 July 2013; and DBA-TRAN-RE-14, the emergency directive on the flight-control computers of the Airbus A320 family of November 2025. The class also cites DBA-TRAN-RE-8, DBA-CHEM-RE-3 and DBA-CHEM-RE-4. All three instruments were written. DBA-TRAN-RE-14 is withdrawn on the author's triage ruling of 9 October 2026 and its designator retired: its F-5 and R-5, on who held the grounding decision, are carried by DBA-TRAN-RE-12 F-6 and R-6, which Section 5.5 takes up, and Conveyances and Fleet records the withdrawal at its v0.2. DBA-TRAN-RE-12 is held by Project 11; DBA-TRAN-RE-13 stays in Project 9's folders until it moves.

From Traffic Management, Signaling and Control, five events are conferred, on the author's choice of 8 October 2026: DBA-TRAN-RE-15, the midair collision over the Potomac River near Ronald Reagan Washington National Airport of 29 January 2025; DBA-TRAN-RE-16, the losses of radar and communications at the Philadelphia approach control facility serving Newark, from April 2025 to September 2026; DBA-TRAN-RE-17, the collision at Chatsworth, California, of 12 September 2008; DBA-TRAN-RE-18, the collision at Fort Totten on the Washington Metro of 22 June 2009; and DBA-TRAN-RE-19, the unauthorized radio-stop broadcasts on the Polish railway near Szczecin of 25 August 2023. The class also cites DBA-TRAN-RE-8 and DBA-CHEM-RE-3. All five instruments were written, and they stay in Project 9's folders until they move. Of the nineteen conferrals, sixteen are live, two are withdrawn and one is recast.

References

[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; DBA-MA Parent Framework (IAEA L1); DBA Series Gaps and Recommendations (§1.7, MA applicability); DBA-CHEM (hazmat boundary and shared release model); DBA-COMM (PNT dependency); DBA-EN (pipeline cross-reference; RT_HVLLC long-lead analog). Regulatory: DOT modal administrations (FAA, FRA, FHWA, FMCSA, MARAD); USCG and USACE (maritime / waterways); TSA (security); PHMSA (hazmat); NTSB (investigation). Class instruments: DBA-TRAN-ND-FC1 (Nodes and Terminals); DBA-TRAN-WY-FC2 (Ways and Linehaul); DBA-TRAN-CV-FC3 (Conveyances and Fleet); DBA-TRAN-CntlCo (Traffic Management, Signaling and Control). Boundary instruments: DBA-DAMS_Sector_Framework (Dams Sector Framework); DBA-CHEM-TR-FC2 (Bulk Chemical Transport). Reference events conferred at Section 9.1: DBA-TRAN-RE-1 to DBA-TRAN-RE-19, sixteen live, held by Project 11 or, until they move, in Project 9's folders; DBA-TRAN-RE-7 and DBA-TRAN-RE-14 withdrawn and DBA-TRAN-RE-11 recast, kept by Project 9. Cross-series instruments cited: DBA-COMM-RE-12 (Baltic Navigation Interference), DBA-COMM-PN-FC4 (Positioning, Navigation and Timing) and DBA-CHEM-RE-5 (West Fertilizer Explosion).

Revision History

0.11 October 2026 — T. Roxey. Records the triage of the series' reference events, their move to Project 11, and the class conformance passes of 10 October 2026. Section 9.1 records all nineteen instruments as written and the live ones as held by Project 11 as they move under the determination of 9 October 2026, the conferral staying here. On the author's rulings of 9 and 10 October 2026 it withdraws DBA-TRAN-RE-7 (the Collins Aerospace attack) and DBA-TRAN-RE-14 (the Airbus A320 directive), records the recast of DBA-TRAN-RE-11 (the Kerch Strait bridge) as an illustration in Ways and Linehaul §5.5, retires the three designators, and records the author's rulings on the Red Sea and Hormuz designator and on the threshold without denial. Section 5.5 takes up DBA-TRAN-RE-12 R-6, which carries the A320 directive's R-5: the authorities of the states of design, registry and operation each hold the decision to ground, each recording its basis under Indication Integrity, the manufacturer's assessment carrying it where it is every authority's basis and the order of the state of design being the act others adopt. Section 6.2 states the consequence of regional satellite-navigation interference for aviation and maritime navigation, on DBA-COMM-RE-12 as Project 11's batch 10 note routes it, and why a node's or a way's power requirement reaches the utility as quality of service rather than through the water series' elevation rule (Project 9 OI-27, joined to OI-17). Section 7.5 corrects the dates of the Middle Eastern airspace closure to the European Union Aviation Safety Agency's bulletins CZIB-2026-03 and CZIB-2026-07, as Project 11's batch 20 note records, and adds the people and conveyances held inside an enclosed sea, on the Red Sea and Hormuz instrument's F-5 as that note refers it. Section 9 records the elements the node and ways instruments return to this framework, to be taken in at its next conformance pass, and the author's decision on use cases. An independent check of the revision against the notes, the instruments and the classes raised 10 items, all taken before issue, the notice to Project 11 being given the same day. Project 11 is given notice of this revision. Revision-history entries are left as records.

0.10 October 2026 — T. Roxey. Sections 4, 5 and 7 are written from the four class instruments, on the author's decisions of 8 October 2026, and are no longer skeletons. Section 4 states the design-basis principles; adopts five sector principles the classes found (conditionality, independence shown as built, the fallback measured, common mode counted as one, and acting before the hazard arrives); organizes the envelope in twelve event categories across the classes; states how the eight class measures compose, with stranded flow over time as the common measure of throughput harm; and summarizes the critical functions. Section 5 states the dual design basis for the sector: placement per action in four places, the automated proceed as the placed act, advance discharge and the governed act for changes that reach every unit, who holds the Command Broker role, autonomous conveyances, what a supplier declares, the two Lines and the governmental acts outside the design basis, and continuity; the skeleton's sentence on automated movements is removed. Section 7, retitled The Military-Action Extension, consolidates the presumptions of applicability, applies the four characteristics to the sector, states the governing campaign as the four classes' scenarios together, treats foreign chokepoints and airspace as imported conditions, adopts the campaign principle for the design duration and leaves the number to a decision across series, gathers the declared residuals, and sets re-determination

triggers. Section 9 records the developed sections and the open duration. An independent check of the sections against the four classes and their sources raised 17 items, 17 taken. Revision-history entries are left as records.

0.9 October 2026 — T. Roxey. The register enters DBA-TRAN-CntlCo (Traffic Management, Signaling and Control) at working-draft status, the series' coordination class, so that all four classes are issued. Section 3.5 records the author's decisions of 8 October 2026 on the class: capacity restoration time and authority reach as the governing measures; closures split by legal authority; the detector alarm enforced as a stop with the dispatcher alone authorizing resumption; authority for an autonomous conveyance only inside a pre-approved envelope, with a qualified human for any grant above the line; and governmental emergency acts on movement recorded outside the design basis. Section 5 records the decision that an automated grant of proceed is the placed act, with advance discharge only behind a gatekeeper that knows occupancy independently. Section 7 records the presumption of applicability for the key control facilities and the governing scenario. Section 9 restates the open items and records that Sections 4, 5 and 7 are now owed. Section 9.1 confers DBA-TRAN-RE-15 to DBA-TRAN-RE-19 on the five reference events the class proposed. References name the class and the conferred events. Revision-history entries are left as records.

0.8 October 2026 — T. Roxey. The register enters DBA-TRAN-CV-FC3 (Conveyances and Fleet) at working-draft status, the series' third facility class, so that all three physical classes are issued and the coordination class alone is at Anticipated status. Section 3.4 records the author's decisions of 8 October 2026 on the class: the line with DBA-CHEM between the container and the vehicle, the wayside detectors homed in this class, fleet unavailability and control recovery margin as the governing measures, onboard autonomy in this class and the authority to move in the coordination class, and the weaponized conveyance as a design-basis event; Section 6.3 points to the container line. Section 7 records the presumption of applicability for the mobilizable fleet and the governing scenario. Section 9 restates the open items and adds the home of fuel releases in transport. Section 9.1 confers DBA-TRAN-RE-12 to DBA-TRAN-RE-14 on the three reference events the class proposed. References name the class, the chemical boundary instrument and the conferred events. Revision-history entries are left as records.

0.7 October 2026 — T. Roxey. The register enters DBA-TRAN-WY-FC2 (Ways and Linehaul) at working-draft status, the series' second facility class. Section 1.5 states the line with the Dams series at navigation locks, drawn by function on the author's decision of 8 October 2026, and Section 6.3 points to it. Section 3.3 records the author's other decisions of that date on the class: airspace a way only where a physical hazard or a hostile threat denies it; a foreign chokepoint an imported condition; severance time, in its repair and replacement parts, and corridor stranded flow as the governing measures; and a span over navigable water carrying both of the ways it severs, with the vessel strike a required event. Section 7 records the presumption of applicability for ways on the designated defense networks, for ways without an absorbing alternative and for foreign chokepoints, and the governing scenario. Section 9 restates the open items. Section 9.1

confers DBA-TRAN-RE-8 to DBA-TRAN-RE-11 on the four reference events the class proposed. References name the class, the Dams boundary and the conferred events. Revision-history entries are left as records.

0.6 October 2026 — T. Roxey. The register enters DBA-TRAN-ND-FC1 (Nodes and Terminals) at working-draft status, the series' first facility class. Section 3.2 records the author's decisions of 8 October 2026 on the class: the node bounded at its fence, the throughput and occupancy chains with stranded flow and clearance time as their measures, the shared-platform event, operate-through power known to the serving utility, the release-and-network rule at the node, and the line with Commercial Facilities by who acts; and records that the classes are written in register order, which settles the Phase-A selection. Section 7 records the presumption of applicability for strategic seaports and for nodes without an absorbing alternative. Section 9 restates the open items. New Section 9.1 confers DBA-TRAN-RE-1 to DBA-TRAN-RE-7 on the seven reference events the class proposed. References name the class and the conferred events. Revision-history entries are left as records.

0.5 October 2026 — T. Roxey. Section 3.5 states the author's decision of 5 October 2026: when a hazardous shipment is in trouble, the Command Broker role for decisions about its movement is held within this sector's coordination class, and the shipping company owes its knowledge of the cargo to whoever decides about the cargo itself, as DBA-CHEM-CorpCo states from the shipper's side. Section 9 records OI-19 as closed. Revision-history entries are left as records. No change to the taxonomy, the designators or the register.

0.4 October 2026 — T. Roxey. Section 1.2 and Section 1.5 now say that the Energy series governs the pipeline transport of fuels, and that pipelines carrying chemicals other than fuels are governed by DBA-CHEM-TR-FC2 (Bulk Chemical Transport) on the author's decision of 5 October 2026. Section 3.1 no longer calls nodes, ways and conveyances physical-function classes: they are function classes, each realized as a physical facility class, under the author's determination of 30 September 2026. Section 9 records that the shared chemical-release model is held at DBA-CHEM-PR-FC1 and adopted by DBA-CHEM-TR-FC2 together with this framework's boundary rule, and it adds the carrier–shipper division of Command Broker authority as an open item (OI-19). Revision-history entries are left as records. No change to the taxonomy, the designators or the register.

0.3 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 in its placement sentence, and states that determinism decides scope and not placement (P9-11, wording). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.2 August 2026 — T. Roxey. Designators ratified (ND / WY / CV + CntlCo; mode as sub-index; CntlCo reused from Electric as the shared mandated-external-control model).

0.1 August 2026 — T. Roxey. Initial skeleton. Established DBA-TRAN as a standalone sector series; adopted the function-primary decomposition (Nodes / Ways / Conveyances / CntlCo) with mode as the sub-index, and stated the transpose relationship to the conventional modal view (checked against the CIKR sector mindmap, which organizes mode-first). Set the multi-modal DOT-umbrella regulatory inheritance; placed transportation control nearest the electric CntlCo on the coordination spectrum. Closed the DBA-CHEM §3.3 hazmat-conveyance boundary and named the Dams (locks) and Energy (pipeline) interfaces; stated the inward Energy / Communications / PNT dependencies. Registered three physical classes and one functional class at Anticipated, with pipeline cross-referenced to Energy. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.